DNS & DHCP Multi-Domain Infrastructure

Project Domains: Anudip.in / D2216.com

Introduction

This project demonstrates the design of a segmented enterprise network. It connects two separate office locations (Anudip and D2216) through a central router, utilizing automated IP addressing (DHCP) and name resolution services (DNS) to allow seamless communication across different subnets.

Network Topology & Hardware

The network follows a Star-Bus hybrid topology:

- **Router (ISR4331):** Acts as the Gateway (Layer 3) to route traffic between the 192.168.10.0 and 192.168.20.0 networks.
- **Switches (2960):** Provide local connectivity (Layer 2) within each office.
- **Servers (Generic):** Configured as multi-service nodes providing HTTP, DNS, and DHCP.
- **End Devices:** PCs simulating employee workstations.

Theory

A. DHCP (Dynamic Host Configuration Protocol)

DHCP automates the assignment of IP addresses, subnet masks, and gateways.

The DORA Process: 1. Discovery: Client asks "Who is the DHCP server?"

2. Offer: Server offers an IP.
3. Request: Client asks to use that IP.
4. Acknowledgment: Server confirms and sends the Gateway/DNS info.

Role in this project: Servers ensure that PCs in each office automatically receive the correct "Exit Door" (Gateway) IP to reach the other office.

B. DNS (Domain Name System)

DNS translates human-readable names (Anudip.in) into machine-readable IP addresses (192.168.10.2).

A-Records: We used "Address Records" to map the domain names directly to the Server IPs.

Role in this project: Enables cross-office browsing so users don't have to remember complex IP addresses.

C. **Routing & Default Gateway**

A Router connects two different networks. For a PC in the Anudip office to talk to the D2216 office, it sends data to its Default Gateway (the Router's interface). The router then checks its Routing Table to find the correct path to the other subnet.

Configuration Details

IP Addressing Table

Device	**Interface**	**IP Address**	**Subnet Mask**
Router0 (Anudip side)	Gig0/0/0	192.168.10.1	255.255.255.0
Router0 (D2216 side)	Gig0/0/1	192.168.20.1	255.255.255.0
Anudip Server	Fa0	192.168.10.2	255.255.255.0
D2216 Server	Fa0	192.168.20.2	255.255.255.0

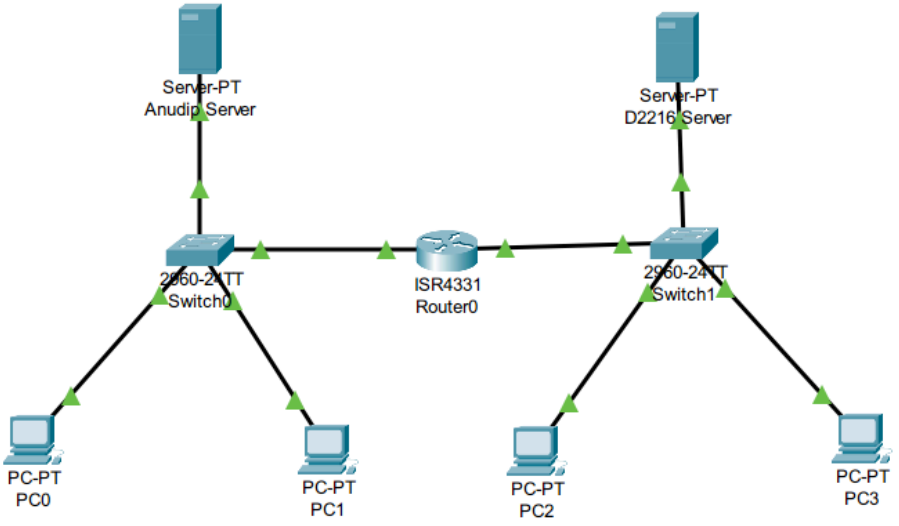
Service Configurations

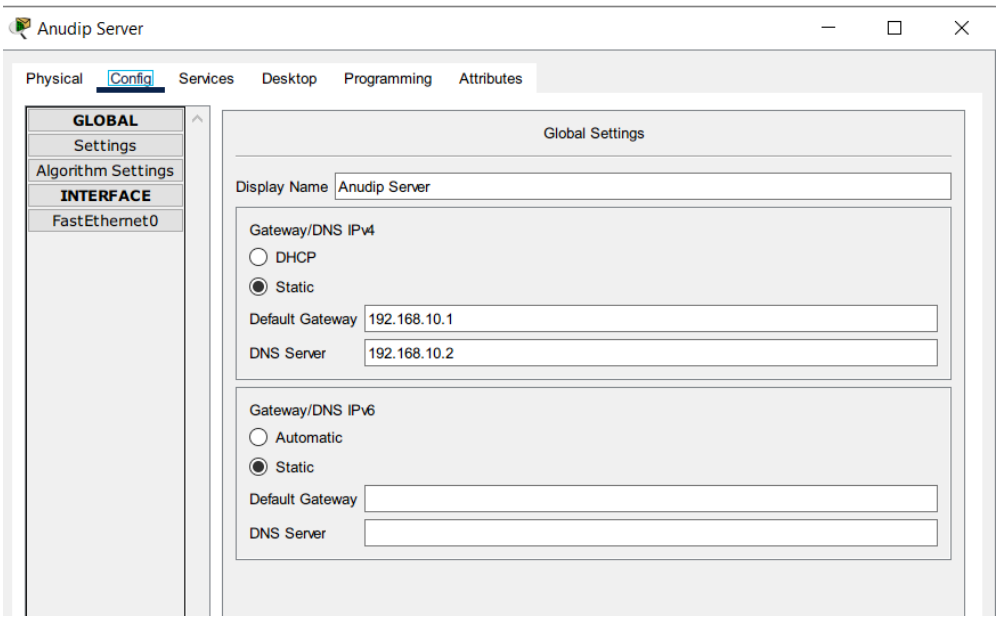
- DNS: Each server holds records for both domains to ensure name resolution works globally.
- DHCP Pools: Configured with specific start addresses (.10) to avoid conflicts with static server IPs.
- HTTP: Customized index.html on each server to provide visual verification of the connection.

Implementation

- **Physical Layer:** Devices connected via Copper Straight-Through cables.
- **Gateway Activation:** Router interfaces enabled via CLI using no shutdown.
- **Static Assignment:** Servers given static IPs to prevent DNS/DHCP failure.
- **Service Setup:** DHCP pools and DNS A-records created.
- **Verification:** Used ipconfig /renew on PCs to pull new settings and ping to test cross-network reachability.

Output:





Anudip Server

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES: HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP

DHCP

Interface: FastEthernet0 — Service: On / Off

Pool Name: anudipPool

Default Gateway: 192.168.10.1

DNS Server: 192.168.10.2

Start IP Address : 192 | 168 | 10 | 10

Subnet Mask: 255 | 255 | 255 | 0

Maximum Number of Users : 246

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add | Save | Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.10.1	192.168.10.2	192.168.10.0	255.255.2...	256	0.0.0.0	0.0.0.0
anudipPool	192.168.10.1	192.168.10.2	192.168.1...	255.255.2...	246	0.0.0.0	0.0.0.0

Anudip Server

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES: HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP

DNS

DNS Service: On / Off

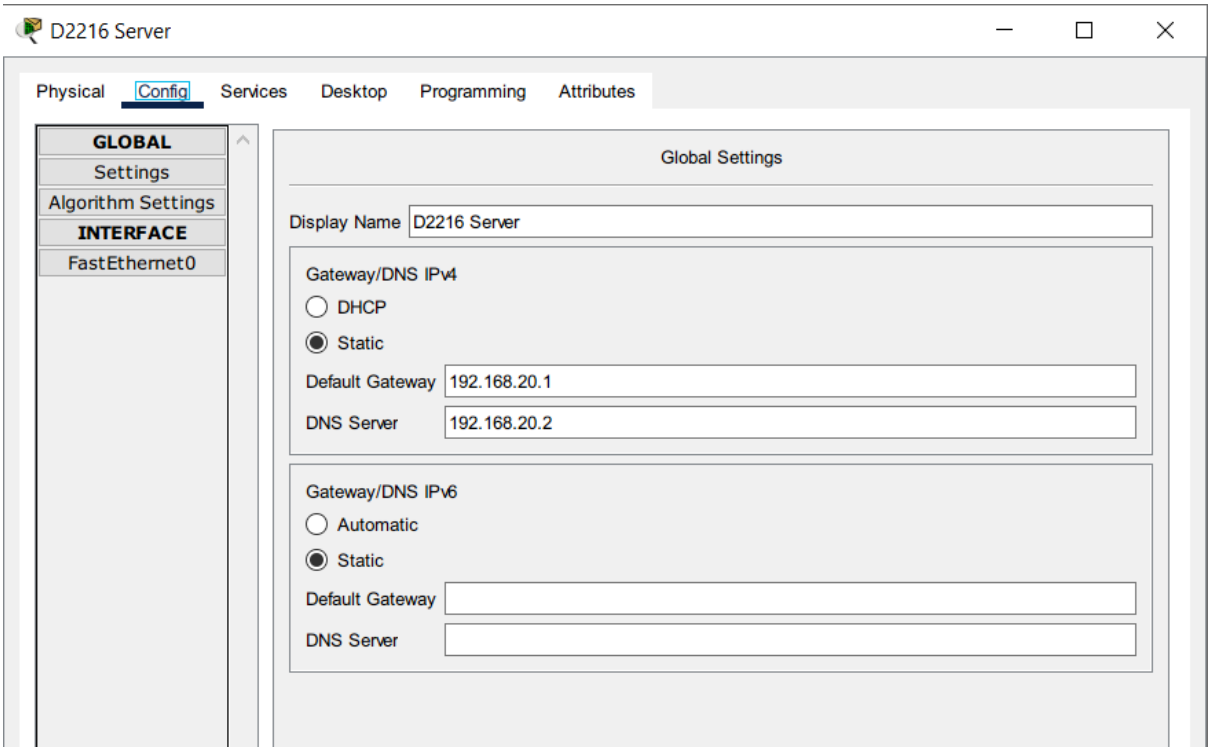
Resource Records

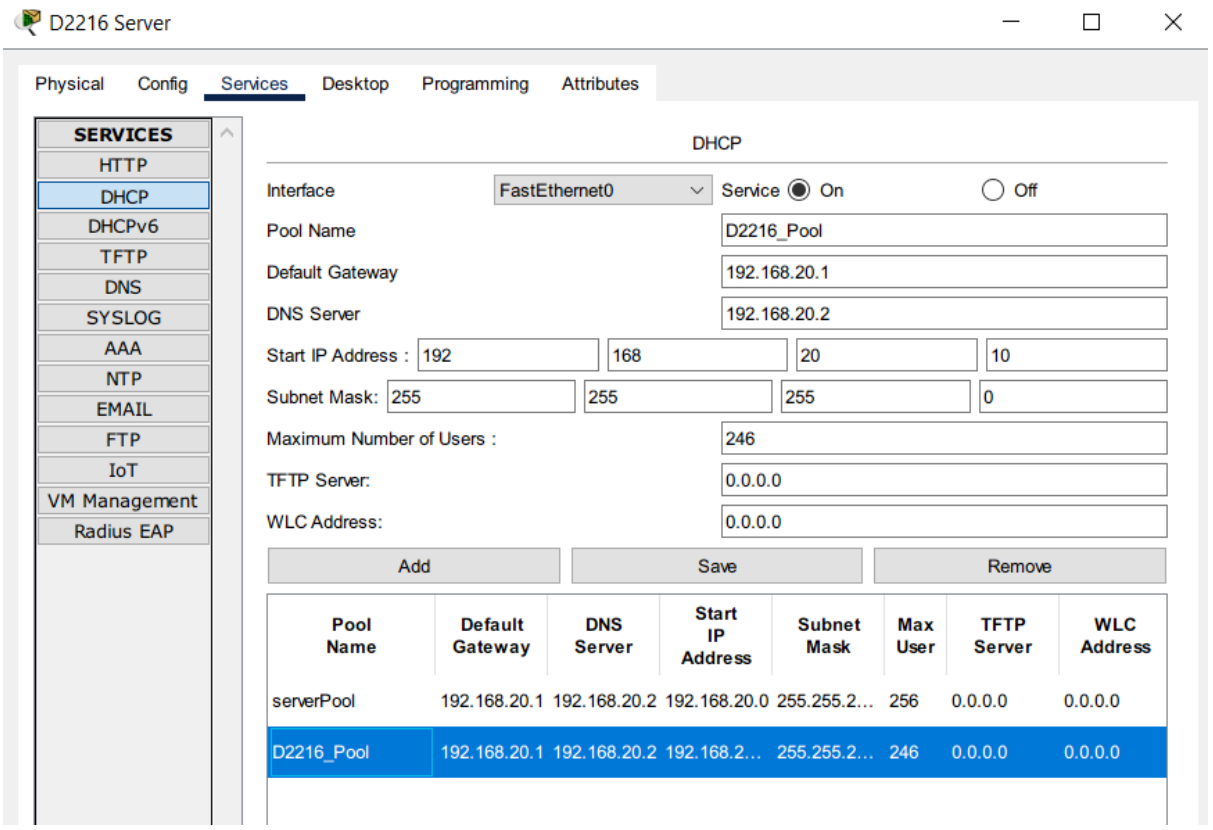
Name: Type: A Record

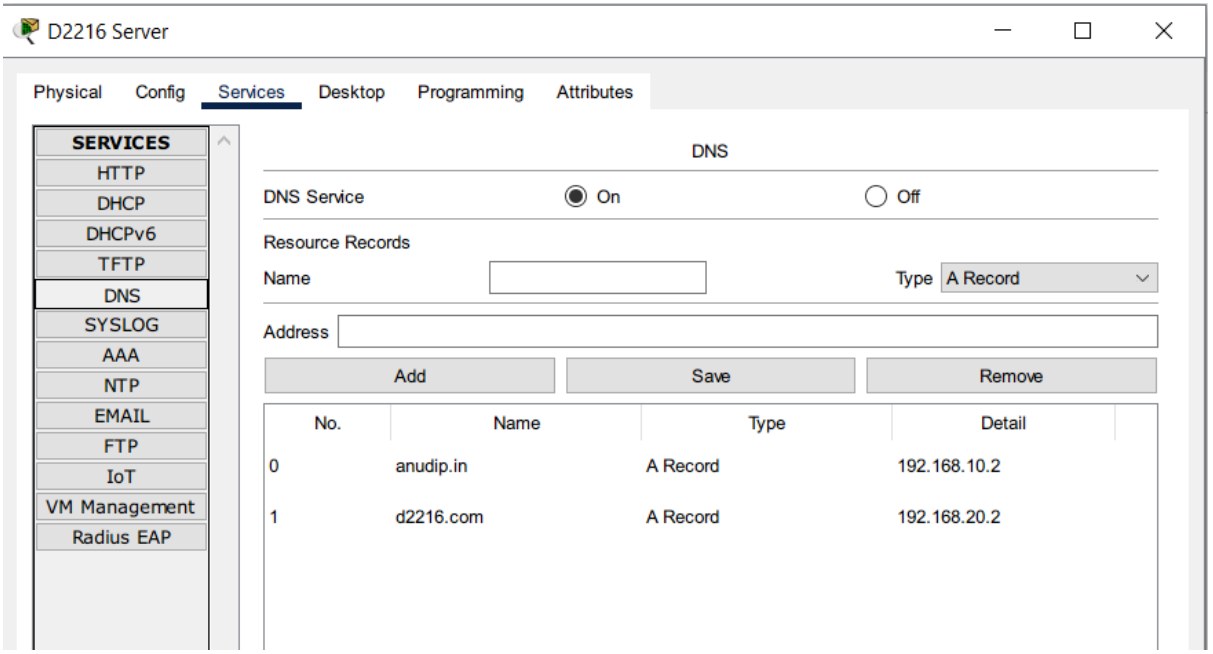
Address:

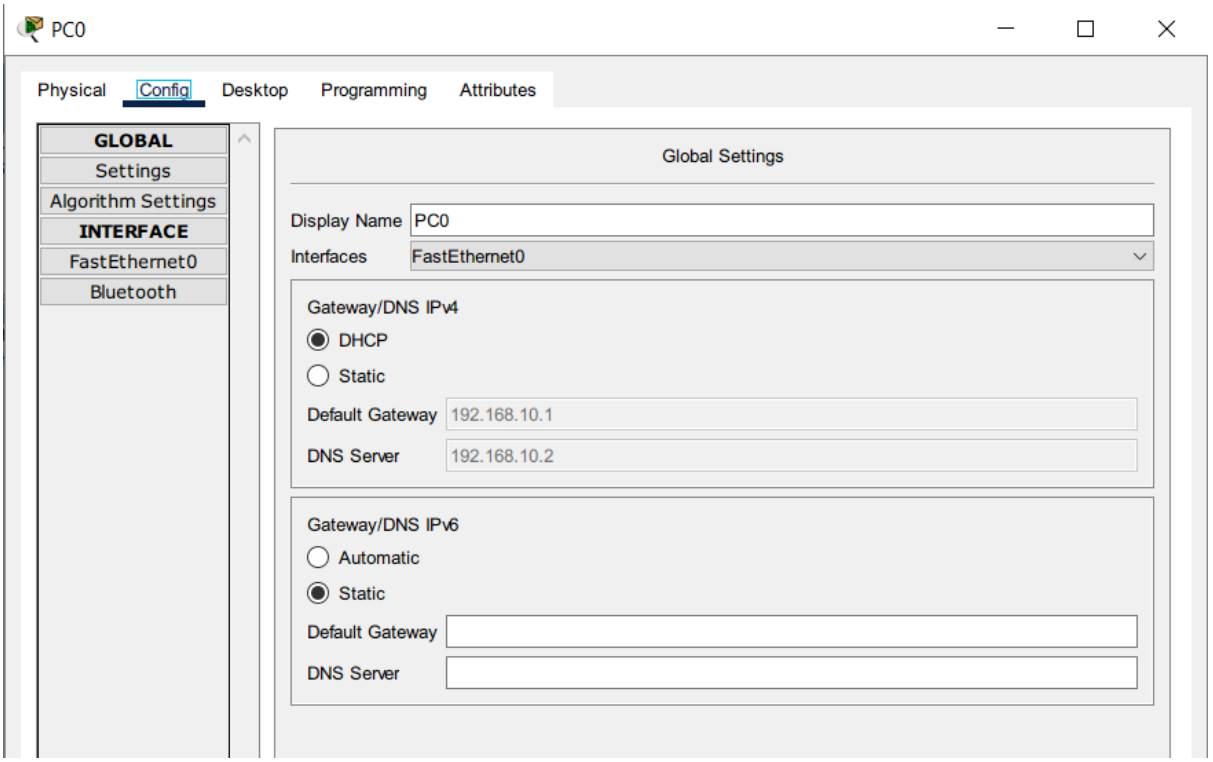
Add | Save | Remove

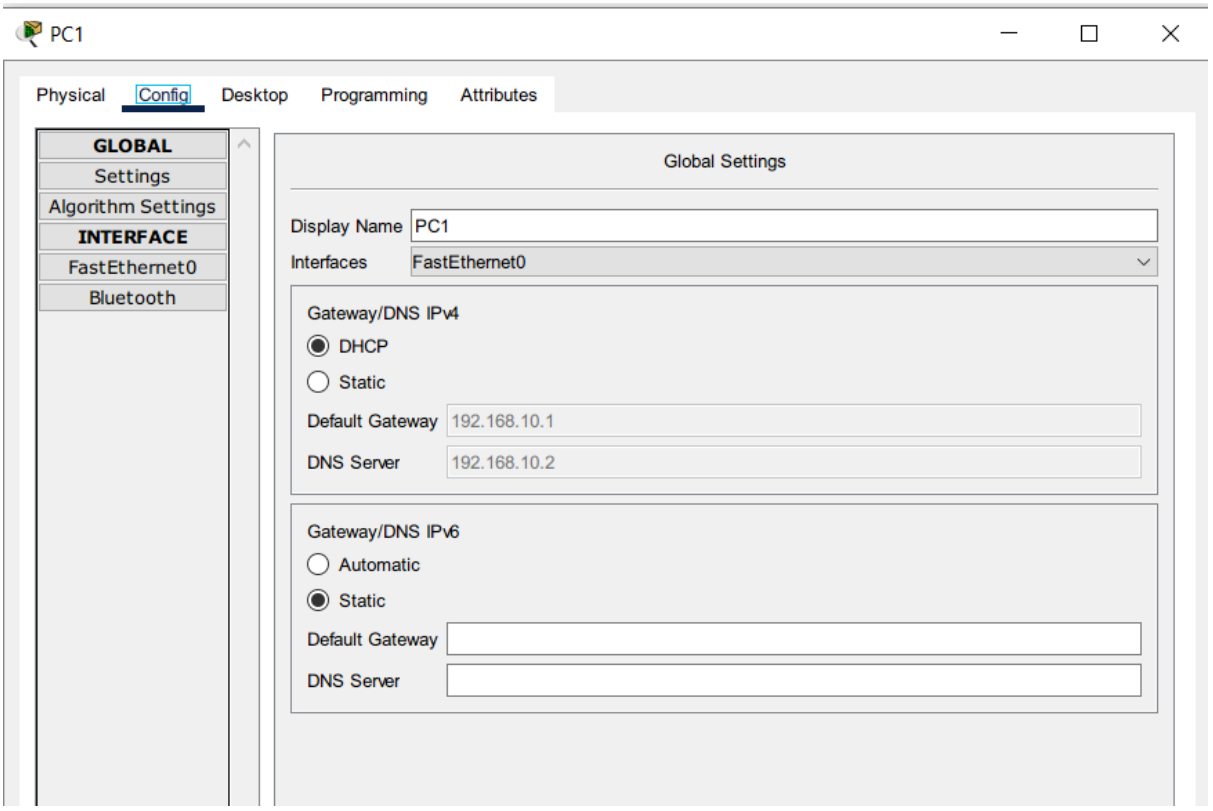
No.	Name	Type	Detail
0	anudip.in	A Record	192.168.10.2
1	d2216.com	A Record	192.168.20.2

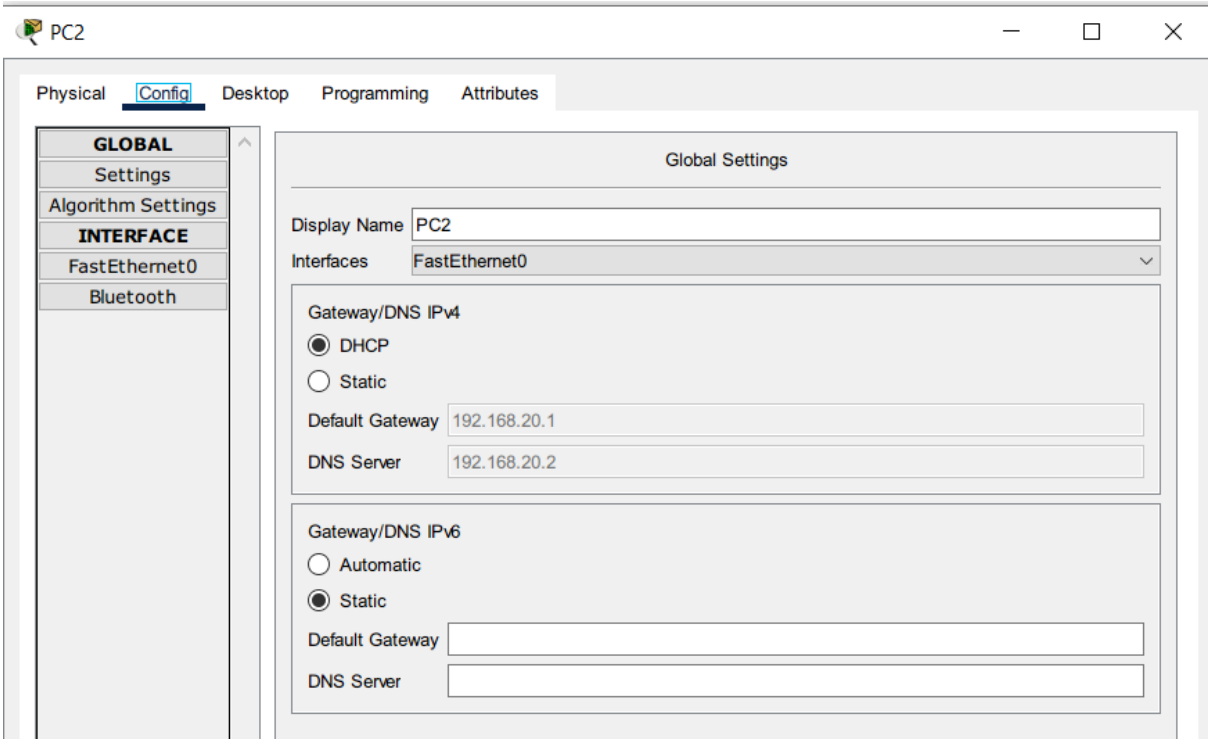

D2216 Server
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name
D2216 Server
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.20.1
DNS Server
192.168.20.2
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server

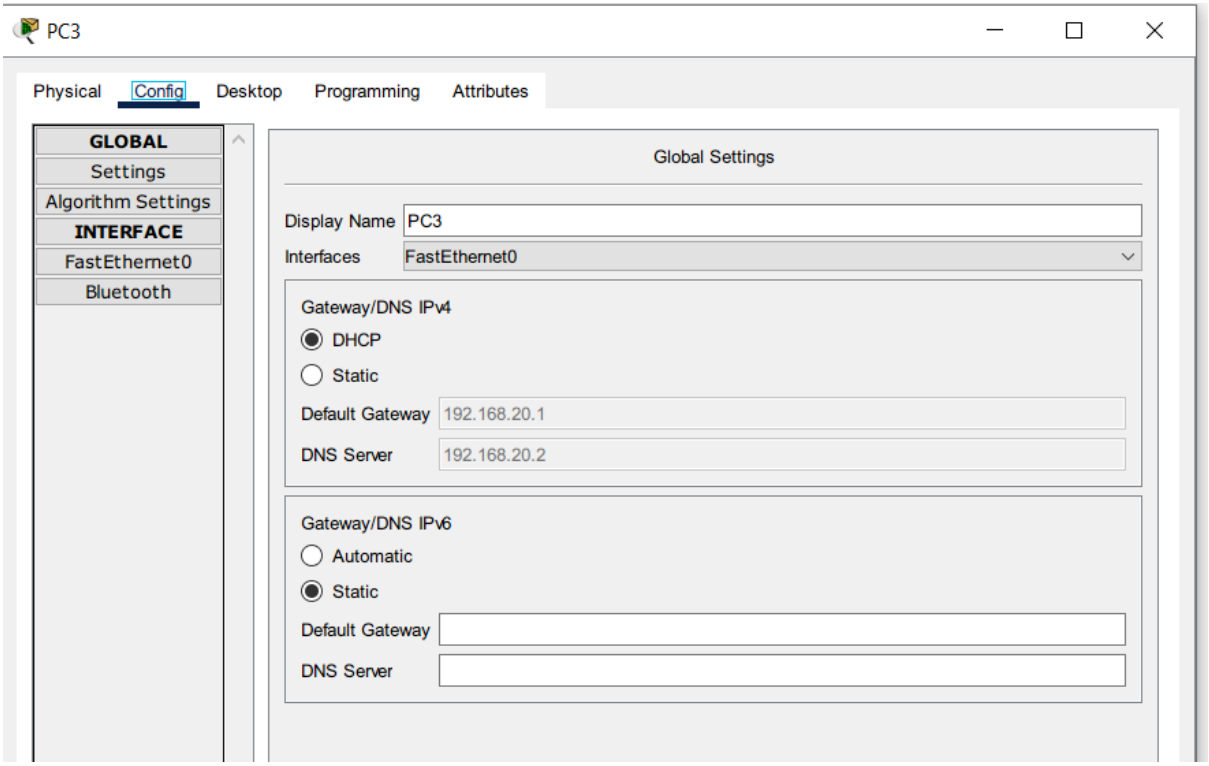

D2216 Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
D2216_Pool
Default Gateway
192.168.20.1
DNS Server
192.168.20.2
Start IP Address :
192
168
20
10
Subnet Mask:
255
255
255
0
Maximum Number of Users :
246
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
192.168.20.1
192.168.20.2
192.168.20.0
255.255.2...
256
0.0.0.0
0.0.0.0
D2216_Pool
192.168.20.1
192.168.20.2
192.168.2...
255.255.2...
246
0.0.0.0
0.0.0.0

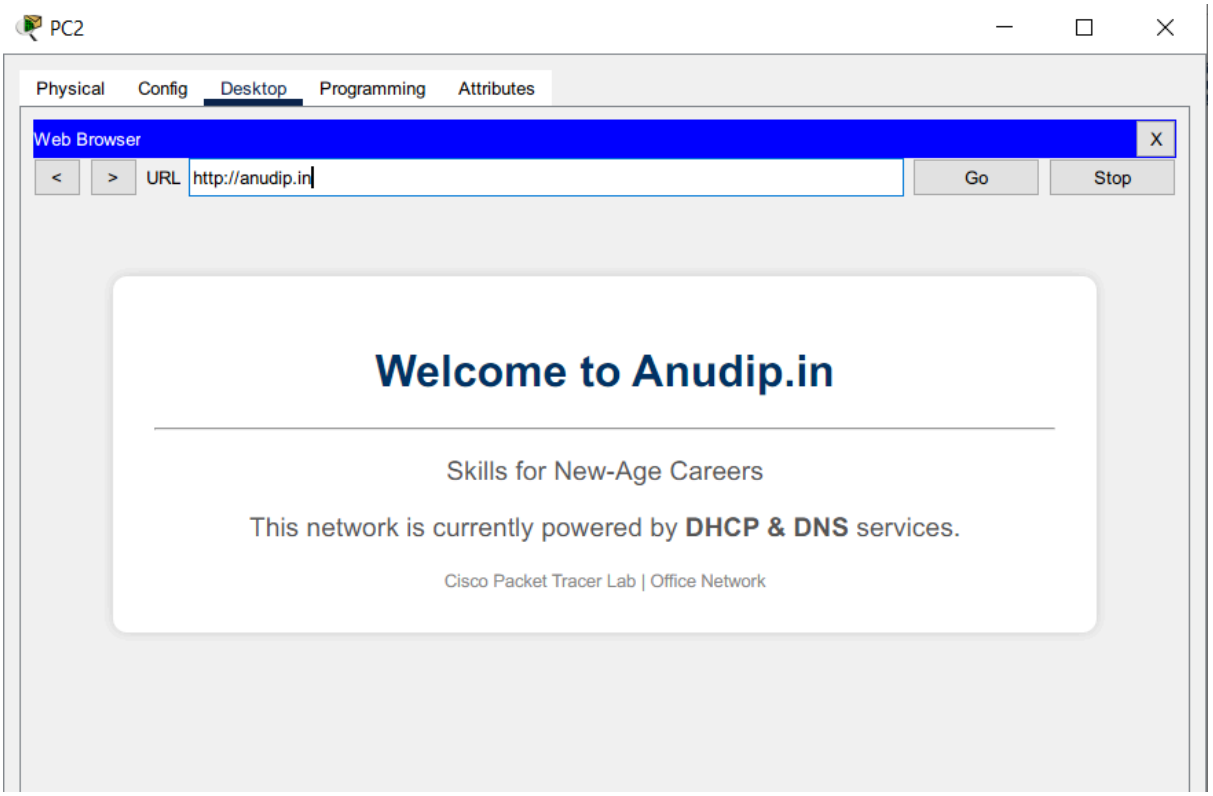

D2216 Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
anudip.in
A Record
192.168.10.2
1
d2216.com
A Record
192.168.20.2

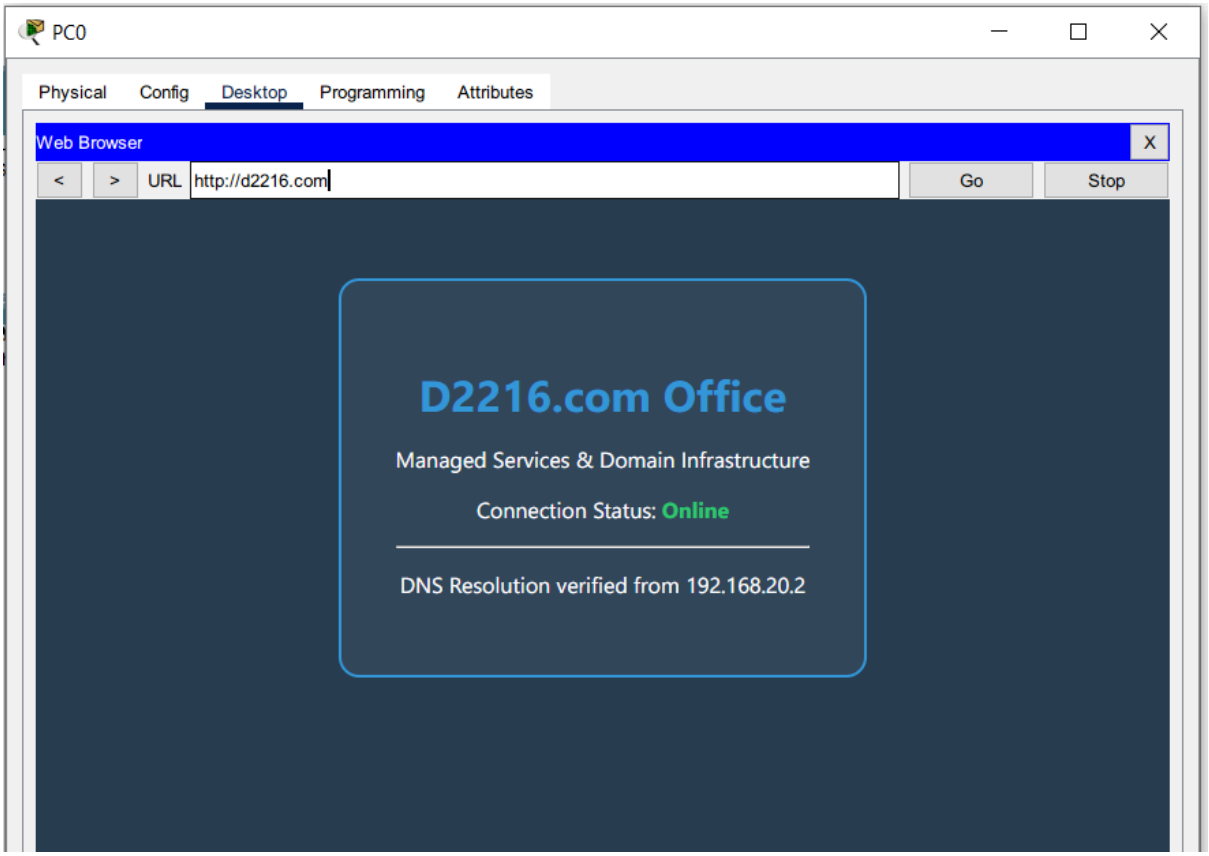

PC0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC0
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.10.1
DNS Server
192.168.10.2
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server


PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC1
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.10.1
DNS Server
192.168.10.2
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server


PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC2
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.20.1
DNS Server
192.168.20.2
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server


PC3
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC3
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.20.1
DNS Server
192.168.20.2
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server


PC2
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://anudip.in
Go
Stop
Welcome to Anudip.in
Skills for New-Age Careers
This network is currently powered by DHCP & DNS services.
Cisco Packet Tracer Lab | Office Network



Conclusion

The project successfully achieves a routed environment where end-users can access web services using domain names. It highlights the importance of the Default Gateway in inter-network communication and the efficiency of DHCP in managing large-scale IP assignments.